

Discrete Mathematical structure (CIBCAIT3)

Unit 1:-

- ① The foundations:- Logic and proofs:- Propositional Logic, Applications of propositional Logic, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers, Rules of Inference, Introduction to proofs, proof Methods and strategy.
- ② Basic structures:- sets, Functions, Sequence, Sums and Matrices:- Sets, set operations, Functions, Sequences and summations, matrices.

Unit 2:-

- ① Counting:- Basics of counting, Pigeonhole principle, Permutation and combination, Binomial coefficient and combinations, Generating Permutation and combination.
- ② Advanced counting Techniques:- Applications of Recurrence relations, solving linear recurrence, Relations, Divide and conquer algorithms and Recurrence Relations, generating functions, Inclusion-Exclusion, Applications of Inclusion.

Unit 3:-

- ① Induction and Recursion:- Mathematical Induction, strong Induction and well ordering, Recursive Definitions and structural Induction, Recursive Algorithms, Program corrections.
- ② Relation:- Properties of relation, closure operation on relation, Equivalence relation and partition, Operation on relation, Representing relation, Composition of relation.

Unit 4:

- ① Graphs : graphs and graph models, Graph Terminology and special Types of Graphs, Representing Graphs and graph Isomorphism, connectivity, Euler and Hamilton Paths, shortest - Path Problems, Planar graphs, Graph coloring
- ② Trees : Introduction to Trees, Application of Trees, Tree traversal, spanning trees, Min. spanning trees

Trees & Spanning → 13/11/25

14/11/25 - Rooted trees

15/11/25 - Relation

Foundation

Advanced Spanning

Induction Division

17/11/25

Problems